Khwaja Yunus Ali University

School of Science and Engineering

Department of Computer Science and Engineering

Final Examination: Fall Semester 2019

Program: B.Sc. in CSE (Batch: 8^{th}) 1^{st} Year 2^{nd} Semester

Course Title: Physics – II

Course Code: PHY 1201

Total Time: 2 hours | Full Marks: 40

Part-B

[Answer the question No. 1 and any four (04) from the rest] **5 × 7 = 35**

1. See the stem given below and answer the following questions:

Work function of sodium and potassium are $5.3ev$ and $2.2ev$ respectively. UV light of 3.5×10^{-7} m wavelength incident on the both metal surfaces.

a) Define Work function. 1

b) Find the energy of incident light. 3

c) Is it possible to happen the photo-electric effect from both metal? – Justify your answer. 2

2. a) Explain the term "Radioactivity never ends". 2

b) The disintegration constant of a radioactive element is 0.0693 per month. Calculate the time required for 75% at the element to decay. 4

3. a) State the Huygens' principle. 1

b) How to form Newton's rings? 3

c) Sketch the interference, diffraction and polarization phenomena of light. 2

4. a) What is inertial frame of reference? 1

b) Explain the theory of relativity is not applicable for the very short velocity of particle compared to the velocity of light. 2

c) At what speed a hypothetical train should move so that its length will be one-third of the stationary length? 3

5. a) Illustrate the Compton scattering. 2

b) If the velocity of proton is $\frac{1}{20}$th of the velocity of light, then calculate the de-Broglie wavelength. 4

6. a) Write down the basic components of nuclear reactor. 1

b) Draw the chain reaction. How is the chain reaction in a nuclear reactor controlled? 2

c) Find the mass defect and binding energy of the given nucleus: 3

$$^{7}_{3}Li$$

Where,

Mass of each neutron, $m_n = 1.008665amu$;

Mass each proton, $m_p = 1.007277amu$; and

Mass of the Lithium nucleus, $M_{Li} = 7.016005amu$

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Part-A

Time: 20 min | **[Tick (√) mark to the appropriate one]** | **$0.5 \times 20 = 10$**

1. Polarization used in …………….
   a) bus headlight.
   b) glare-reducing sunglasses.
   c) damaging the hardness of metal.
   d) slit tracking in wood.
   e) mobile communication.

2. What is the angle between electric field and magnetic field in light?
   a) 30^0
   b) 45^0
   c) 60^0
   d) 90^0
   e) 180^0

3. There are …… types of diffraction.
   a) two
   b) three
   c) four
   d) five
   e) six

4. How many phase differences between two points in the wave front?
   a) $\frac{\pi}{4}$
   b) $\frac{\pi}{3}$
   c) $\frac{\pi}{2}$
   d) π
   e) 0

5. Which pair is isotones?
   a) ${}^{15}_{7}\mathrm{N}\ and\ {}^{16}_{8}\mathrm{O}$
   b) ${}^{16}_{8}\mathrm{O}\ and\ {}^{16}_{7}\mathrm{N}$
   c) ${}^{13}_{6}\mathrm{C}\ and\ {}^{14}_{6}\mathrm{C}$
   d) ${}^{2}_{1}\mathrm{H}\ and\ {}^{3}_{1}\mathrm{H}$
   e) ${}^{7}_{4}\mathrm{Be}\ and\ {}^{7}_{3}\mathrm{Li}$

6. What is the radius of Hydrogen nucleus?
   a) 3.7fm
   b) 3.4fm
   c) 2.7fm
   d) 1.5fm
   e) 1.2fm

7. The nuclear reaction: ${}^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + ?$

a) Alpha
b) Beta
c) Gamma
d) X-ray
e) UV-ray

8. Acceleration of inertial frame of reference is …………..

a) zero
b) infinity
c) negative
d) positive
e) fractional

9. how many energy releases in every fission reaction?

a) 37MeV
b) 180MeV
c) 200MeV
d) 931MeV
e) 1337MeV

10. What is the value of the Compton wavelength?

a) $13.6A^0$
b) $10A^0$
c) $3.6A^0$
d) $0.53A^0$
e) $0.024A^0$

11. What is the unit of half-life?

a) sec
b) sec^{-1}
c) min^{-1}
d) $year^{-1}$
e) Bq

12. What is the mass of the protons?

a) $1 \times 10^{-31}kg$
b) $1.6 \times 10^{-19}kg$
c) $6.63 \times 10^{-31}kg$
d) $1.67 \times 10^{-27}kg$
e) $6.23 \times 10^{-23}kg$

13. The …………… is the minimum value of incident light which can eject and electron out of the metal.

a) stopping potential
b) threshold frequency
c) critical wavelength
d) critical intensity
e) threshold potential

14. The slope of the kinetic energy of ejected electron vs frequency of incident photon curve is equal to ………….

a) stopping potential
b) threshold frequency
c) critical wavelength
d) work function
e) Planck constant

15. What is the half-life of Radon?
   a) 2.88days
   b) 3.82days
   c) 14.6days
   d) 28days
   e) 192days

16. Which is the common form of nuclear reaction?
   a) a(X,Y)b
   b) b(Y,X)a
   c) a(X)b
   d) X(a,b)Y
   e) Y(X,a)b

17. When the velocity v of the body is negligibly small as compared to light velocity c, then we can write,
   a) $L < L_0$
   b) $L > L_0$
   c) $L = L_0$
   d) $L < L_0 < 0$
   e) $L > L_0 > 0$

18. What is the unit of work function?
   a) J
   b) N
   c) Hz
   d) M
   e) Volt

19. Fusion occurs in …………
   a) Atom bomb
   b) Hydrogen bomb
   c) Nuclear reactor
   d) Particle accelerator
   e) Isotope production process

20. Which one is the alpha particle?
   a) ${}^{1}_{1}H$
   b) ${}^{2}_{1}H$
   c) ${}^{3}_{1}H$
   d) ${}^{1}_{0}n$
   e) ${}^{4}_{2}He$

Invigilator Signature

Examiner Signature